AC-Powered LED Lamps Without Rectifiers

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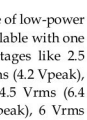
Usually, LED lamps require rectifiers when connected to the AC mains power supply. Electrical isolation of the LED lamps from the mains is also required in most cases. But rectifiers create a switching noise and also add to their cost. Presented here is a simple circuit for LED lamps that doesn't require the use of rectifiers.

Test Points

Test point	Details
Across TP0 and TP1	Around 4.5V AC
Across TP2 and TP3	Around 230V AC

There is a wide range of low-power mains transformers available with one or more secondary voltages like 2.5 Vrms (3.5 Vpeak), 3 Vrms (4.2 Vpeak), 3.3 Vrms (4.7 Vpeak), 4.5 Vrms (6.4 Vpeak), 5 Vrms (7.1 Vpeak), 6 Vrms (8.5 Vpeak), 6.3 Vrms (8.9 Vpeak) and with power rating from 1.3 VA up to 12 VA. The primary current of these transformers is relatively low—around 15 to 80 mA—and, when needed, can be limited with resistors without producing a lot of heat.

Most of these transformers are PCB-mounted and easy to use. On the other hand, LED lamps with power rating of 1 W to 10 W are good enough to produce light for your workstation and for a small room. Most LEDs today have forward drop voltage of 1.6 V to 3.7 V depending on the technology and the colour. The working current for these LEDs is usually 20 mA to 100 mA and sometimes higher. The parameters of the LEDs match very well with the parameters of small 2.5Vrms-6.3Vrms transformers mentioned above.



Circuit and working

This LED lamp with transformer does not need rectifiers. The circuit uses the internal resistance of the transformer and low-cost protecting resistors to limit the current through the LEDs. The LED lamp with four levels of power is described here.

Fig. 1 shows the circuit of an LED lamp using transformer X1 with secondary voltage of 4.5 V. Resistor R1 limits the voltage and current peaks over transformer X1. Resistor R2 and capacitor

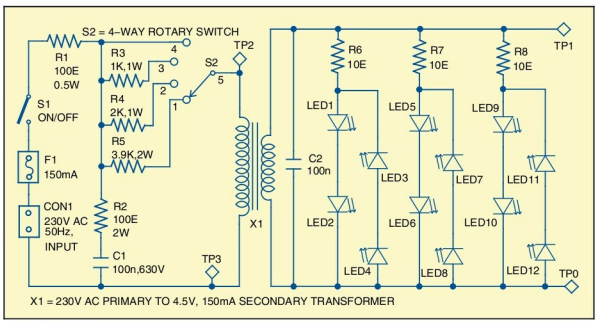


Fig. 1: Circuit of an LED lamp

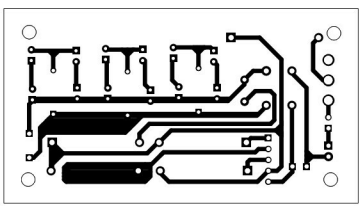


Fig. 2: An actual-size, single-side PCB for the AC-powered lamp

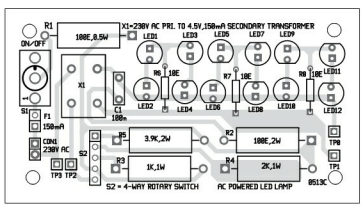


Fig. 3: Component layout for the PCB